

with handhelds or laptops.

The problem, however, is that our current WIMP system was built for mice and keyboards, so it's not ideal for touchscreens.

In our June 2007 issue, we spoke about Jeff Han's multi-touch interface, which allows you to use multiple points of contact, or even have multiple people using the same interface at the same time. Han's interface is revolutionary: it will allow users to actually get a feel for applications that needs graphics, zooming and interaction. But such an interface is useless for, say, gaming, because there's no interaction required. Ditto for word processing. However, let's say you're managing or editing photographs, browsing the Net, or using Google Earth. That's where Han's vision comes in. Perhaps the way gaming has broken away into the specially-made machines we know as consoles today, the future might see different hardware for different purposes.

Another example here is Microsoft's Surface, a 30-inch multi-touchscreen that's kept horizontally instead of vertically, and is embedded in a desk. It allows multiple users to use it at once. It uses Windows Vista, a PC embedded in the desk, and five cameras to sense motion. It can sense 52 simultaneous touches, computing them and translating them into on-screen inputs.

Check out Jeff Han's work at <http://cs.nyu.edu/~jhan/>, and the Microsoft Surface at <http://www.microsoft.com/surface/>.

Direct Interaction

You've come across Natural Language Processing (NLP) and Brain-Computer Interface (BCI) in this space. We've discussed them at length earlier, so we won't get into the details, but here's about two major technologies that promise to get us interfacing better with computers.

NLP: NLP is the ability of a computer to understand and compute a human's normal speaking voice. The perfect NLP system would allow you to talk to your computer the way you would talk to another person. We're far from achieving this—subtleties of voice patterns, background noises, accents, and more play spoilsport (refer *Tech Transcends Tongues*, *Digit*, April 2006.) Of course that doesn't stop us from trying, and there are a lot of NLP projects being worked upon: check out <http://opennlp.sourceforge.net/projects.html>, a list of open source projects related to NLP.

BCI: Imagine a computer that could read your mind—yes, your *mind*—there's no talking required. Imagine sitting in a restaurant talking to your laptop... now you know why NLP isn't really ideal for the way we currently use computers. However, a computer that can read your thoughts... now



We basically have to un-teach people what they have learned so far about computing, and convince them that they can use several fingers, that several people can work on the screen at once"

Jeff Han
Researcher
New York University



The much-hyped Surface takes boring computing to mind-bogglingly amazing new levels

that's something we all want. Imagine thinking out that letter, moving your character in a game just by thinking about it... even better, imagine having a random thought, "I wonder who does the voice of Bart Simpson?" while interfaced with your PC.

Currently, BCI is being researched by quite a few groups all over the world. Most notable is Fraunhofer's Berlin Brain-Computer Interface (BBCI; more at http://ida.first.fraunhofer.de/bbci/index_en.html), which is already showing off the ability for people to play *Pong* against each other or the computer, using their thoughts as controls. Fraunhofer also holds BCI-related contests to encourage the development of this technology. Currently, accuracy for such a system in ideal conditions is between 91 and 99 per cent, which is extremely good—and leaves us optimistic.

The Future

It will take years before we have decent prototypes (leave alone products). We are also faced with the challenge of making a billion people forget all they know about computers, and re-learn to use them from scratch. Technologies such as eye-controlling the mouse, which uses a camera to monitor your eye and translate it into data your PC can understand is all very well and good, but what use if we still have to type out our letters?

Brain-Computer Interfaces also have limitations: how do we stop the PC from trying to interpret thoughts we don't want it to? Imagine working in an office, looking over to that hot co-worker and thinking, "I wish I could send him / her an e-mail asking him / her out," then turning back to see that your PC just shot off a mail asking the boss out!

New GUIs for operating systems / software and new hardware bring us to the chicken-and-egg scenario—software waiting for new input devices to code for, and devices waiting to see what software or AI can make possible.

We loved the concept of Microsoft's WinFS, and were looking forward to having a file system that was built like a database. Unfortunately, looks like we won't be seeing anything like it for PCs soon. Still, we'll keep our fingers crossed. WinFS promised to be different from the WIMP interface, and perhaps just that excitement of seeing something new has us hooked.

Another trend we have to consider is the *divergence* factor. When PDAs were killed by smartphones, *Digit* told you all about Convergence—the coming together of gadgets, hardware, software, and services into one do-it-all gadget. Things are slowly beginning to shift in the opposite direction. Sure, we all want more powerful cell phones, and who wouldn't want a device powerful enough to run Vista yet small enough to fit in one's pocket, and which you can use to make and receive phone calls? Still, how do we explain gaming consoles and their runaway popularity? Then there are the Internet tablets that have been popping up, allowing you limited functionality as compared to a PC—while still being ideal for browsing!

Perhaps the future of BCI is planting a chip in the brain—who knows? Apologies for the corniness when we tell you that all we can do is wait and watch. ☒

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